Part 5: Prompt Engineering Challenge

Below are five prompt scenarios demonstrating effective prompt engineering techniques, including the prompt used, the AI's response, and an explanation of the prompt's effectiveness.

Scenario 1: Summarising Notes

The Prompt I Wrote:

> ***"Act as an expert science tutor. Summarize the following text about photosynthesis into exactly 3 concise bullet points suitable for a 7th-grade student to understand easily: [Insert text about plants converting sunlight, water, and CO2 into glucose and oxygen]."***

The Full AI Response:

> - Plants act like tiny solar-powered factories, using sunlight to make their own food.
> - They take in carbon dioxide from the air and water from the ground to create a type of sugar for energy.
> - As a bonus, this process releases fresh oxygen into the air, which humans and animals need to breathe!

Why This Prompt Worked:

> This prompt was highly effective because it used **role-playing** ("Act as an expert science tutor") to set the tone. It also defined the **target audience** ("7th-grade student"), ensuring the vocabulary was simple, and provided a strict **formatting constraint** ("exactly 3 concise bullet points") so the output wouldn't be overly long or dense.

Scenario 2: Generating a Study Plan

The Prompt I Wrote:

> ***"I have a beginner Python programming exam in 7 days. I have exactly 2 hours available to study each evening. Create a day-by-day study plan covering variables, loops, functions, and data structures. Include a mix of theory and practical coding each day."***

The Full AI Response:

> Here is your 7-day Python study plan (2 hours/day):
>
> - **Day 1: Variables & Data Types** (1 hr reading theory/syntax, 1 hr practicing basic print statements and variable assignments).
> - **Day 2: Conditional Logic** (1 hr theory on if/elif/else statements, 1 hr writing small logic programs like an age checker).
> - **Day 3: Loops** (1 hr learning 'for' and 'while' loops, 1 hr practicing iterating through ranges).
> - **Day 4: Functions** (1 hr understanding def, parameters, and return values, 1 hr writing custom calculators).
> - **Day 5: Basic Data Structures** (1 hr learning Lists and Dictionaries, 1 hr coding data storage exercises).
> - **Day 6: Mini-Project Integration** (2 hrs building a simple text-based game or contact book combining all concepts).
> - **Day 7: Review & Mock Test** (1 hr reviewing weak spots, 1 hr doing practice exam questions without looking at notes).

Why This Prompt Worked:

> This prompt worked because it provided specific **context** (beginner Python exam) and clear **limitations** (7 days, 2 hours per day). By explicitly listing the topics to cover and requesting a specific format ("mix of theory and practical coding"), it forced the AI to generate a realistic and highly actionable schedule rather than generic advice.

Scenario 3: Creating Interview Questions

The Prompt I Wrote:

> ***"Act as a hiring manager for a tech company. Generate 3 behavioral interview questions for a junior front-end developer position. Focus specifically on teamwork and problem-solving skills. For each question, briefly describe what a 'good' answer should include."***

The Full AI Response:

1. **Question:** *"Tell me about a time you disagreed with a team member on a design or coding choice. How did you handle it?"*
   What to look for: A good answer will show empathy, a willingness to communicate openly, and a focus on data or user experience over personal ego to reach a compromise.
2. **Question:** *"Describe a situation where you couldn't find the bug in your code. What steps did you take to resolve it?"*
   What to look for: Look for logical debugging steps, knowing when to take a break, using tools (like console.log or debugger), and recognizing when it is time to ask a senior developer for help.
3. **Question:** *"How do you handle working on a task when the requirements provided by the product manager are vague?"*
   What to look for: A strong candidate will emphasize proactive communication, asking clarifying questions, and perhaps creating a quick prototype to align expectations before diving deep into coding.

Why This Prompt Worked:

> Assigning a persona ("hiring manager") gave the output a professional tone. Specifying the exact job role and the specific soft skills to target (teamwork, problem-solving) kept the questions highly relevant. Asking for an evaluation rubric ("what a good answer should include") made the output immediately useful for an actual interviewer.

Scenario 4: Writing a Resume Summary

The Prompt I Wrote:

> ***"Write a professional resume summary for a recent computer science graduate. I have 6 months of internship experience in web development using HTML, CSS, and React. I am passionate about building accessible user interfaces. Keep the summary under 50 words and use an energetic but professional tone."***

The Full AI Response:

> "Detail-oriented Computer Science graduate with internship experience in front-end web development. Proficient in HTML, CSS, and React. Passionate about building inclusive, accessible user interfaces and eager to leverage technical skills to drive impactful web solutions in a collaborative, fast-paced team environment."

Why This Prompt Worked:

> This prompt succeeded because it supplied all the necessary **variables** upfront: education level, specific experience time, tech stack, and personal passion. Furthermore, setting a strict **length constraint** (under 50 words) and specifying the **tone** (energetic but professional) ensured the output was polished, concise, and ready to be pasted directly into a resume.

Scenario 5: Creating a Lesson Plan

The Prompt I Wrote:

> ***"Create a 45-minute lesson plan for high school physics beginners introducing the concept of 'Gravity'. Structure the plan strictly with: a 5-minute warm-up, a 10-minute core lecture, a 20-minute interactive activity, and a 10-minute wrap-up assessment."***

The Full AI Response:

> **Lesson: Introduction to Gravity (45 Minutes)**
>
> **Warm-up (5 mins):** The Drop Test. Drop a heavy book and a crumpled piece of paper simultaneously. Ask the class to observe and hypothesize why they hit the ground at the same time, challenging preconceived notions about weight and falling speed.
>
> **Core Lecture (10 mins):** Direct instruction defining gravity as an attractive force between all masses. Introduce Newton's Law of Universal Gravitation in simple terms, explaining how mass and distance affect gravitational pull.
>
> **Interactive Activity (20 mins):** Gravity Simulation Labs. Have students use tablets to access the PhET "Gravity Force Lab" simulation. Working in pairs, they will adjust the mass of spheres and the distance between them, recording how the gravitational force changes on a guided worksheet.
>
> **Wrap-up Assessment (10 mins):** Exit Ticket. Students must answer three quick questions before leaving: 1) Define gravity. 2) If the mass of an object increases, what happens to its gravitational pull? 3) Explain the difference between mass and weight.

Why This Prompt Worked:

> This prompt was highly effective because it provided a clear **structural template**. By mandating specific time blocks (5 min, 10 min, 20 min, 10 min) and specific phases of the class (warm-up, lecture, activity, assessment), the AI was forced to organize its knowledge into a highly practical and immediately usable document for a teacher.